

Basic Information

Experiment No.	03
Date	17/09/2026
Title	Employee Salary Details using Interface and Runtime Polymorphism

Objective / Aim

To write and execute a Java program to display the salary details of an employee using an interface, implementation classes, and runtime polymorphism. The program determines whether an employee is a Regular or Contract employee based on the first character of the employee ID and displays the corresponding salary details.

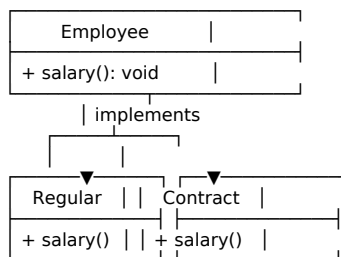
System Requirements

Hardware	Standard PC/Laptop; Intel Core i3/i5 processor or equivalent; minimum 4 GB RAM; minimum 500 MB free storage
Software	Windows / Linux / macOS; Java Development Kit (JDK 17 or JDK 21); Eclipse, IntelliJ IDEA, NetBeans, VS Code, o

Algorithm / Logic Description

1. Create an interface named Employee containing the salary() method.
2. Create a Regular class that implements Employee and define salary() to display regular employee salary details.
3. Create a Contract class that implements Employee and define salary() to display contract employee salary details.
4. Create the EmployeeSalary main class and import java.util.* for Scanner input.
5. Create a Scanner object and accept the employee ID.
6. Declare an Employee reference variable e.
7. Check the first character of the employee ID using id.charAt(0).
8. If the first character is 'R', create a Regular object and assign it to e.
9. Otherwise, create a Contract object and assign it to e.
10. Call e.salary(). The appropriate salary() method is selected at runtime, demonstrating runtime polymorphism.
11. Display the corresponding salary details.

UML Class Diagram



Relationship:

Regular and Contract implement Employee.

Employee reference 'e' refers to different objects at runtime, demonstrating runtime polymorphism.

Relationship: Regular and Contract implement the Employee interface. The Employee reference variable can refer to either implementation object at runtime, and the corresponding salary() method is executed.

Source Code

```
import java.util.*;

interface Employee {
    void salary();
}

class Regular implements Employee {
    public void salary() {
        System.out.printf("Salary Details:\n");
        System.out.printf("Basic Pay: 25000 HRA: 15000 T.A: 5000 Total Amount: 45000");
    }
}

class Contract implements Employee {
    public void salary() {
        System.out.printf("Salary Details:\n");
        System.out.printf("Basic Pay: 12000 HRA: 0 T.A: 3000 Total Amount: 15000");
    }
}

public class EmployeeSalary {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.printf("Enter Employee Id: ");
```

```
String id = sc.next();

Employee e;

if(id.charAt(0) == 'R')
    e = new Regular();
else
    e = new Contract();

e.salary();
}

}
```

Compilation & Execution Commands

Save the source code exactly as EmployeeSalary.java.

```
javac EmployeeSalary.java
java EmployeeSalary
```

Output Screen

```
Enter Employee Id: R101
Salary Details:
Basic Pay: 25000 HRA: 15000 T.A: 5000 Total Amount: 45000
```

Result

The Employee Salary Details program was successfully implemented in Java using an interface and runtime polymorphism. The program accepts an employee ID, identifies the employee as Regular or Contract based on the first character of the ID, and displays the corresponding salary details successfully.